HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • **Stamp 1G — Full-Time Work Eligible**

PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Designed and implemented control logic for **Rockwell PLCs** and integrated with site **SCADA systems** to automate a new batch processing line, reducing manual intervention by **40%**.
- Executed fault finding and repair on automated production equipment, diagnosing and resolving complex issues in PLC code for **Siemens S7 controllers**, restoring operation within critical SLA windows.
- Collaborated with cross-functional teams to develop and deploy new automation features for a **Robotic pick-and-place cell**, enhancing assembly speed by **15%**.
- Provided on-call support for automation systems, responding to and resolving critical issues on production equipment during out-of-hours shifts, maintaining system availability.
- Developed GAMP documentation, including URS and FAT, for a new vision system integration, ensuring compliance with regulatory standards and project requirements.
- Monitored process performance data, identifying opportunities for automation enhancements and defining technical specifications for new control system hardware.
- Containerized legacy control system interfaces using **Docker** and deployed to an **AWS EC2** environment, improving remote access reliability and enabling faster troubleshooting.

SKILLS

PLC/Robotics – Rockwell, Siemens S7, Telemecanique, PLCs, Robotics, Servo Motors, Motion Control

SCADA/HMI – SCADA Systems, HMI

Networking – EtherNet IP, DeviceNet, ControlNet, Modbus, Profibus, Automation Network Protocols

Systems Integration – MES, Control Systems Integration, Vision Systems (Cognex)

Project Management – URS, FDS, FAT, SAT, GAMP, Capital Projects, Budget Management, Requisitions

Compliance/Safety – Machinery Directive, CE Compliance, EN60204-1, GMP, EHS, RCAs

Cloud/Containerization – AWS, EC2, Docker

PROJECTS

Automated Batch Processing Line Control System

- Designed and programmed the control system for a new batch processing line using **Rockwell PLCs** and integrated it with the site's **SCADA** infrastructure.
- Developed fault-tolerant logic to manage servo motor operations and ensure seamless data flow to the **MES**, achieving an estimated **35%** reduction in cycle times.
- Authored detailed GAMP documentation, including URS and FAT protocols, to validate system performance and ensure adherence to GMP standards.

Vision System Integration for Quality Control

- Implemented a **Cognex vision system** for automated product inspection, developing custom logic to identify defects with **98% accuracy** on the production line.
- Integrated the vision system's output with a **Siemens S7 PLC** for real-time reject management and data logging to the central database.
- Collaborated with the manufacturing team to refine inspection parameters and reduce false positive rates by **20%**.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)